

Attachment C. Lay Reader Survey Instrument

Readability of Online Patient Education Materials for Pre-Procedure Cardiac CT, With an Artificial Intelligence

Rewrite Arm and Lay Reader Assessment

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Passages are presented one at a time, in random order. The source website and the identity of any chatbot are removed. Participants are not told which passages are original and which are rewritten. The survey is delivered as an anonymous online link, or on paper for a participant who prefers paper.

Section 1. Screening and background questions

These questions appear once, at the start, immediately after the participant agrees to the information sheet. They confirm eligibility and let us describe the reader group. No name, contact information, or other identifying information is requested.

- Are you 18 years of age or older? (Yes / No). If No, the survey ends with a thank-you message.
- Can you read English? (Yes / No). If No, the survey ends with a thank-you message.
- Do you work in health care, or have you trained as a doctor, nurse, or other health professional? (Yes / No). If Yes, the survey ends with a thank-you message, because this study needs readers who do not work in health care.
- What is your age range? (18–29 / 30–44 / 45–59 / 60–74 / 75 or older / Prefer not to say)
- What is the highest level of education you have completed? (Did not finish high school / High school diploma or equivalent / Some college or vocational training / Bachelor's degree / Graduate or professional degree / Prefer not to say)

Section 2. How to complete the survey

The following instructions are shown to the participant before the first passage.

- You will see a series of short passages about heart scans, one at a time, in a random order.
- You will not be told where a passage came from or whether a computer program helped write it. Please judge each passage only on what you read.
- After each passage you will answer a few short questions about how clear it was and about what it said.
- There are no right or wrong answers for the clarity ratings. We want your honest impression.
- You may skip any question, and you may stop at any time without giving a reason.
- The passages are study material only. They are not medical advice for you.

Section 3. Rating questions asked after every passage

The same three rating questions are asked after each passage. Each is answered on a 1 to 5 scale. The exact wording of each question and of the scale points is shown below.

Rating item (data field)	Question shown to the reader	Response scale (1 to 5)
Understandability lay_r1	How easy was this passage to understand?	1 = Very hard to understand; 2 = Hard; 3 = Neither easy nor hard; 4 = Easy; 5 = Very easy to understand
Clarity of wording lay_r2	How clear was the wording?	1 = Very unclear; 2 = Unclear; 3 = Neither clear nor unclear; 4 = Clear; 5 = Very clear
Confidence to act lay_r3	If this passage were about your own care, how confident would you feel using it to know what to expect or what to do?	1 = Not at all confident; 2 = Slightly confident; 3 = Somewhat confident; 4 = Quite confident; 5 = Very confident

Data mapping. In the master data file (Attachment G) these three ratings are recorded for each reader as lay_r1 (understandability), lay_r2 (clarity of wording), and lay_r3 (confidence to act). Fields lay_r4 and lay_r5 are reserved and remain blank; they are used only if additional rating items are added by an approved amendment.

Section 4. Comprehension questions asked after every passage

After the ratings, each passage is followed by two or three short multiple-choice questions that check what the passage actually said. These are not a test of the reader; they check whether the passage communicated its key points. Each question has one answer that is supported by the passage and always includes a “The passage does not say” option, so a reader is never forced to guess. The number of questions answered correctly is recorded for each reader as lay_comprehension_correct.

Authoring rule, fixed before data collection. For each source passage, two to three questions are written, each drawn from a key fact a patient would need: what the test is, a key preparation or safety instruction, or a stated risk. The questions are written once from the original source passage, and the identical questions are reused for that passage’s rewrites, so that comprehension is compared on exactly the same facts across the original and each rewrite.

Example (illustrative; a coronary CT angiography passage)

Passage shown to the reader: “A coronary CT angiogram uses a special X-ray scanner to take detailed pictures of the arteries that supply the heart. You may be given an iodine-based dye through a small tube in a vein in your arm so the arteries show up clearly. You may also be given a medicine to slow your heart rate so the pictures are clear.”

- Q1. How are the pictures of the heart arteries made? (a) With an X-ray scanner [correct] (b) With sound waves (c) With a magnet (d) The passage does not say
- Q2. How is the dye given? (a) As a tablet you swallow (b) Through a small tube in a vein [correct] (c) By breathing it in (d) The passage does not say
- Q3. Why might you be given a medicine before the scan? (a) To help you sleep (b) To slow your heart rate [correct] (c) To thin your blood (d) The passage does not say

The correct answer, marked here in brackets for the reviewer, is not shown to the reader. Answer order is fixed for a given passage and identical across its original and rewritten versions.

Section 5. Optional comment after every passage

Was anything confusing or unclear in this passage? You may write a short comment. This question is optional and may be left blank. Free-text comments are stored with the ratings and are used only to describe common points of confusion; they contain no identifying information.

Section 6. Closing screen

Thank you for helping with this study. The passages you read were study material only and are not medical advice. If you have questions about your own care, please speak with your own doctor. If you have questions about this study, contact the principal investigator at naeem.muhammad@mayo.edu. If you have questions about your rights as a research participant, contact the Mayo Clinic Institutional Review Board (IRB study number 26-008664) at [Mayo Clinic IRB telephone number and email to be inserted before submission].